Architecture Justification – Campus Placement & Internship Pipeline

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Architecture Selection: We chose Layered Architecture for the Campus Placement & Internship Pipeline System.

Architectural Choice: Layered Architecture with eight components in three layers. Presentation: Student Portal and Officer Console. Business: Application & Status Service (FR-001, FR-005), Drive Rules & Eligibility Service (FR-002), Interview Pipeline Service (FR-003) and Offer Letter Service (FR-004). Data: Data Access & Encryption Component and Relational Database. Each layer calls only the layer below it, through provided and required interfaces such as IApplication, IDriveConfig, IEligibilityCheck, IEligibleList, IRoundOutcome, IDataAccess and IStorage.

Reason 1 – Placement rules change from one drive to the next: every company sets its own CGPA cut-off, backlog limit and branch list, and the Placement Officer edits them through the Officer Console. In this design that logic sits in a single component, the Drive Rules & Eligibility Service, which the Application & Status Service and the Interview Pipeline Service call through interfaces. A rule change therefore needs no change to either screen, the offer process or the database tables.

Reason 2 – The project is a small campus system built by a team of four: the stages are clear (apply, screen, interview, offer) and the users are two groups at one institution. Because the layers and services are separate, each member can build and test one component against its interface without waiting for the others. Microservices would bring separate deployments, service-to-service networking and distributed transactions that this scale does not justify, and a single-server Client-Server design would mix screens, rules and storage together.

Security Advantage: The Data Access & Encryption Component is the only path to storage. The portals can never reach the database, and every business service must go through IDataAccess. Because every read and write passes through this one component, AES-256 encryption at rest for academic records and offer documents (NFR-001) is applied in a single place and cannot be skipped by any other part of the system.

Performance Benefit: The business services keep no session state, so several identical copies can run side by side behind the user interface to handle the peak of 500 concurrent students with page responses under 3 seconds (NFR-002). Students apply against the same few active drives, so the Drive Rules & Eligibility Service can hold those rules in memory and skip a database lookup on most eligibility checks.

Trade-off: Every request crosses layer boundaries, which adds a little call overhead, and one busy service cannot be scaled in complete isolation as it could with Microservices. We accept this in return for simpler development, clear responsibilities and one controlled path to the data.